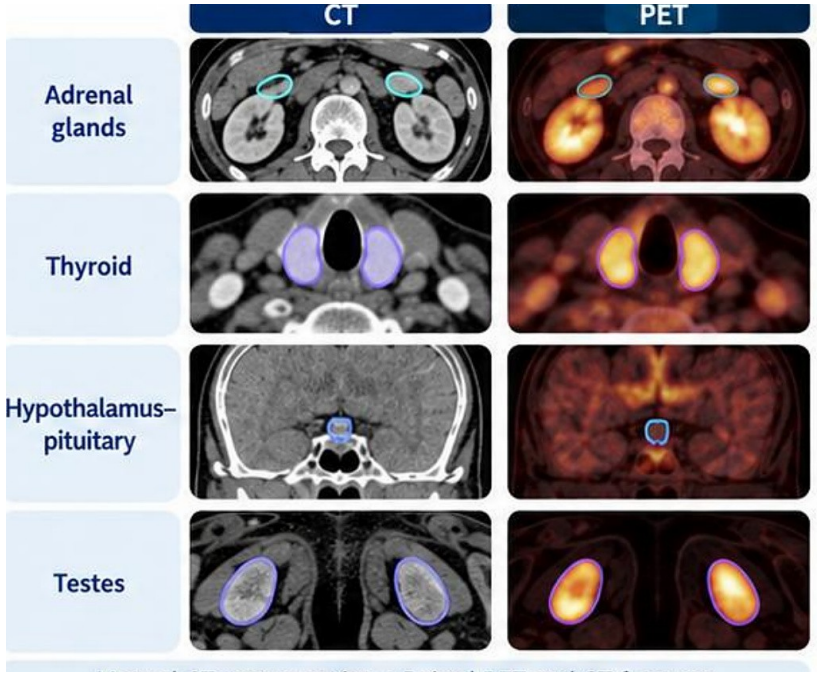
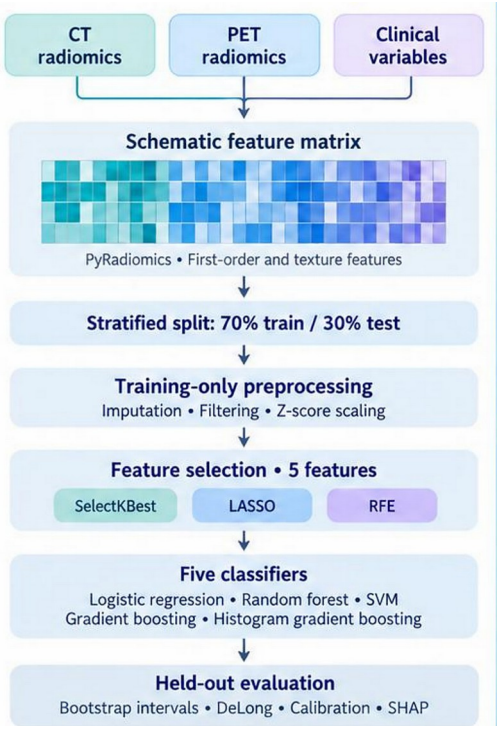
RADIOMICS · SYSTEMIC PHENOTYPING

Endocrine-organ PET/CT radiomics



Normal endocrine organs on PSMA PET/CT



CT + PET + clinical fusion model

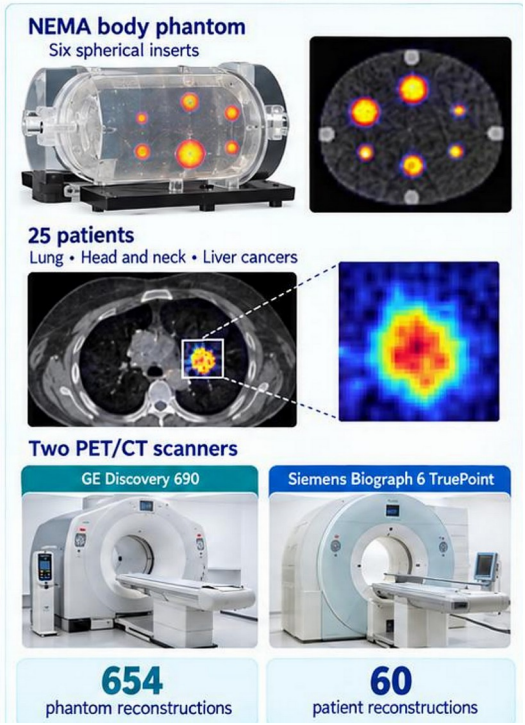
Key points

- 101 men after prostatectomy
- Organs beyond the tumour carry signal
- Fusion model adds modest value

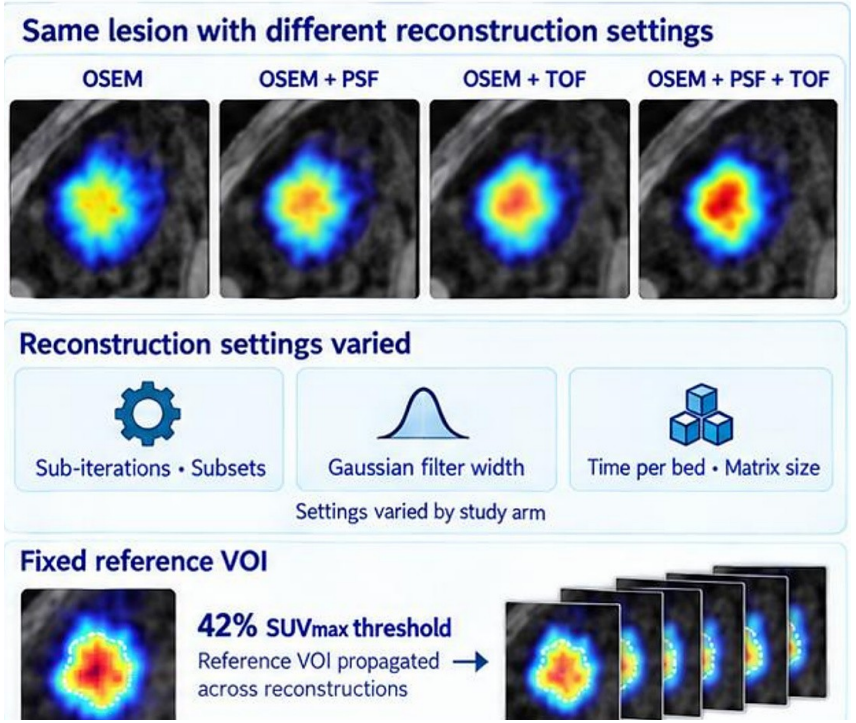
Systemic imaging signatures beyond visible lesions

RADIOMICS · ROBUSTNESS

PET reconstruction changes radiomic features



Phantom + patients, 2 scanners



Same lesion, many reconstructions

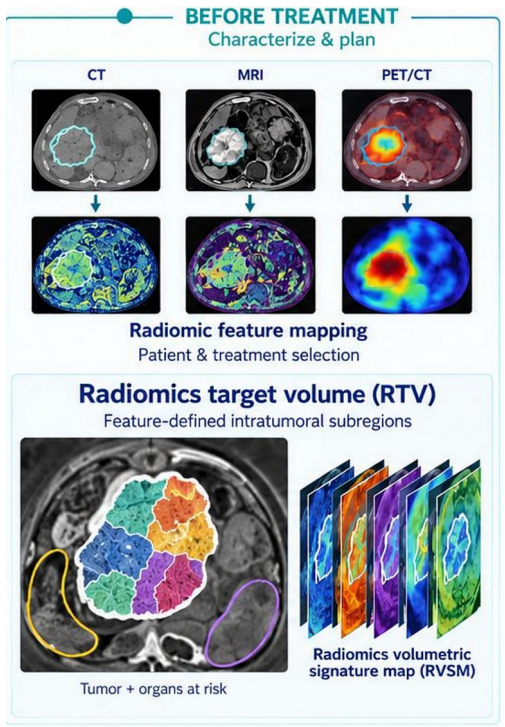
Key points

- About half of features stay stable
- Texture features most sensitive
- Matrix size has the largest effect

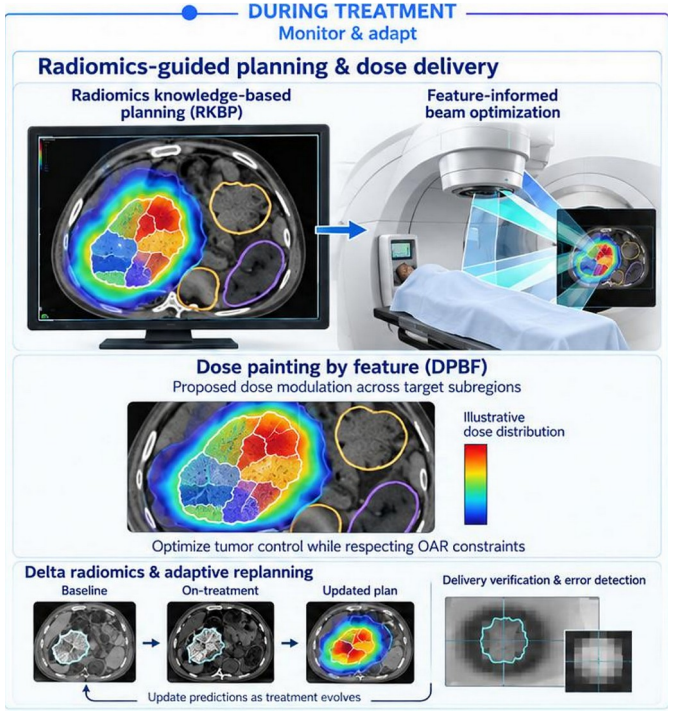
Standardize reconstruction before using radiomics

RADIOMICS-GUIDED RADIATION THERAPY

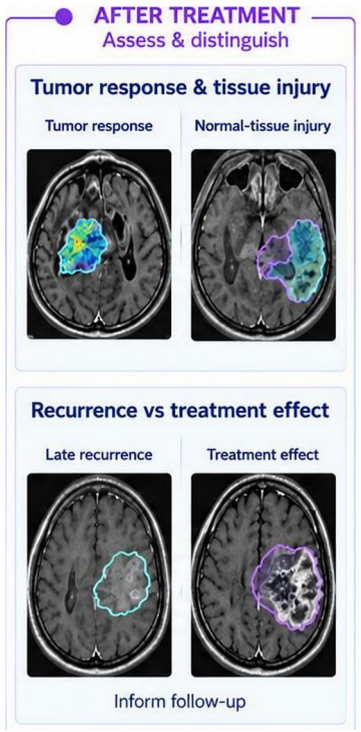
Radiomics across the radiotherapy timeline



Before: characterize and plan



During: monitor and adapt

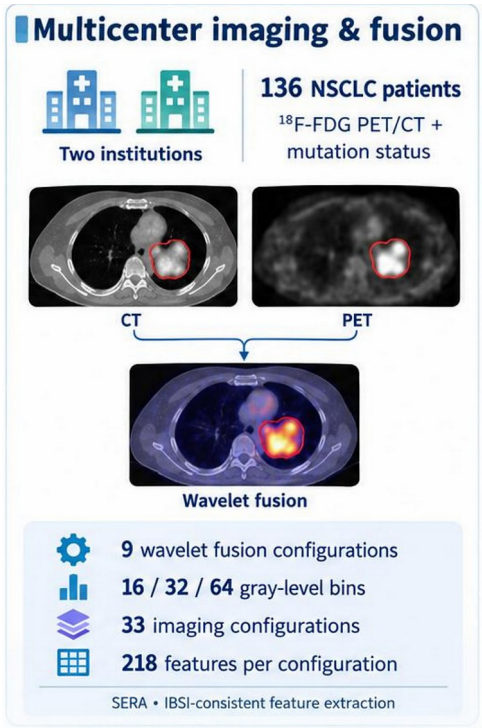


After: assess response

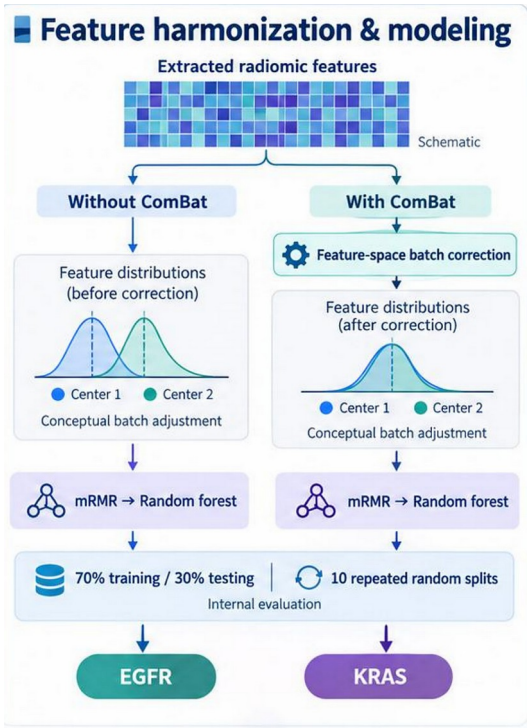
From feature maps to dose painting and adaptive replanning

RADIOGENOMICS · MULTICENTRE HARMONIZATION

ComBat harmonization for EGFR and KRAS



136 NSCLC, two centres, PET/CT fusion



Models with and without ComBat

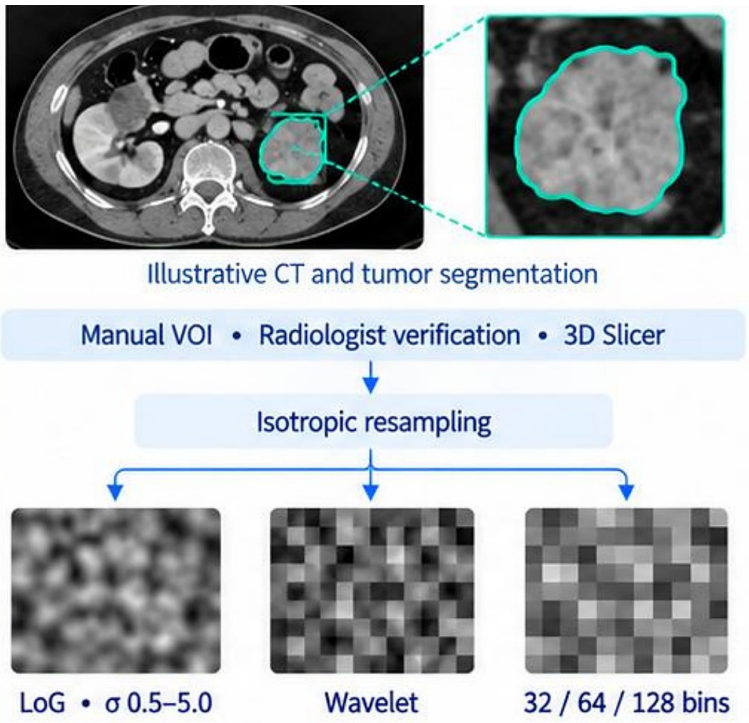
Key points

- Centre effects distort features
- ComBat removes batch differences
- Better EGFR and KRAS prediction

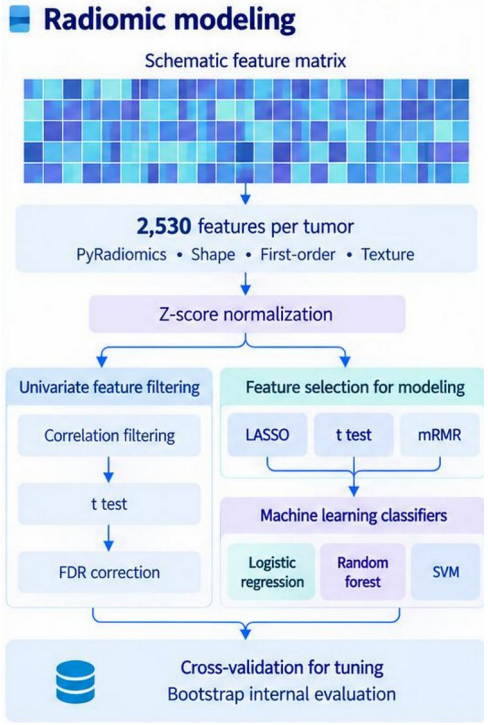
Harmonization makes multicentre radiomics more reliable

RADIOMICS · TUMOUR GRADING

CT radiomics for Fuhrman grade in ccRCC



71 ccRCC CT scans, filtered images



2,530 features → machine learning

Key points

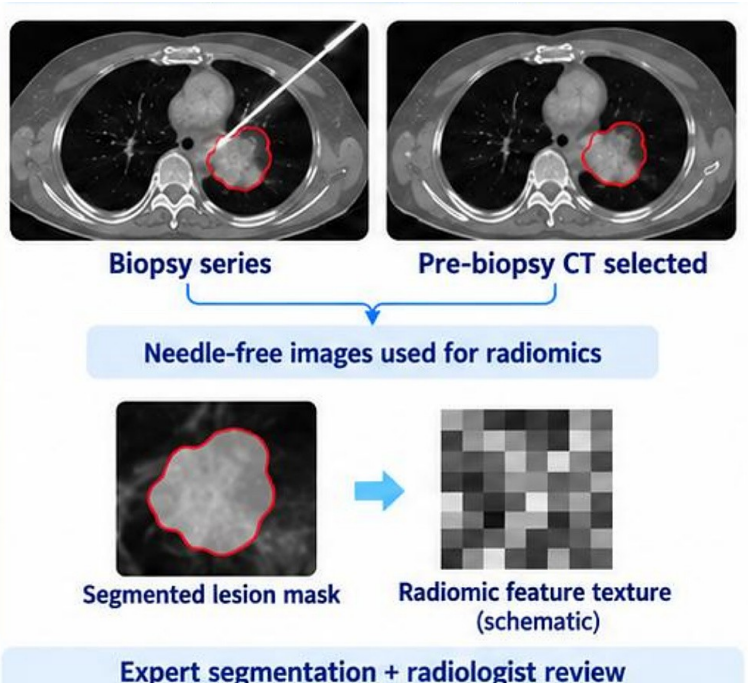
- Grading without surgery
- Filtered (LoG) images work best
- Exploratory; needs validation

Potential non-invasive preoperative grading

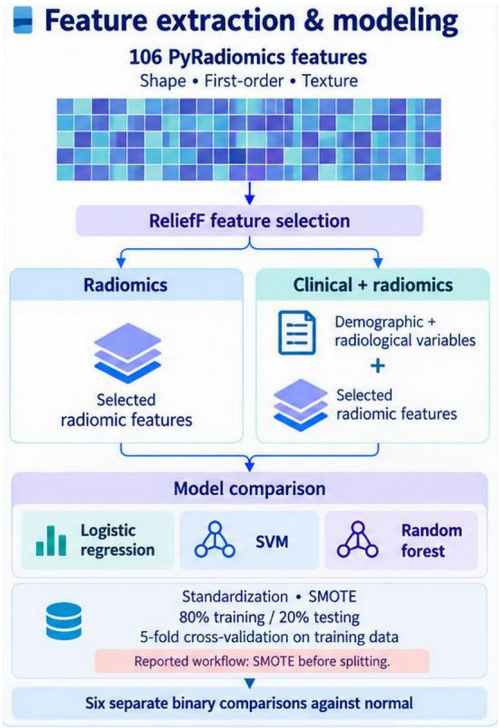
Nazari et al., La Radiologia Medica 2020

RADIOMICS · VIRTUAL BIOPSY

CT-based virtual biopsy of thoracic lesions



Pre-biopsy CT, 300 patients



Feature selection + three classifiers

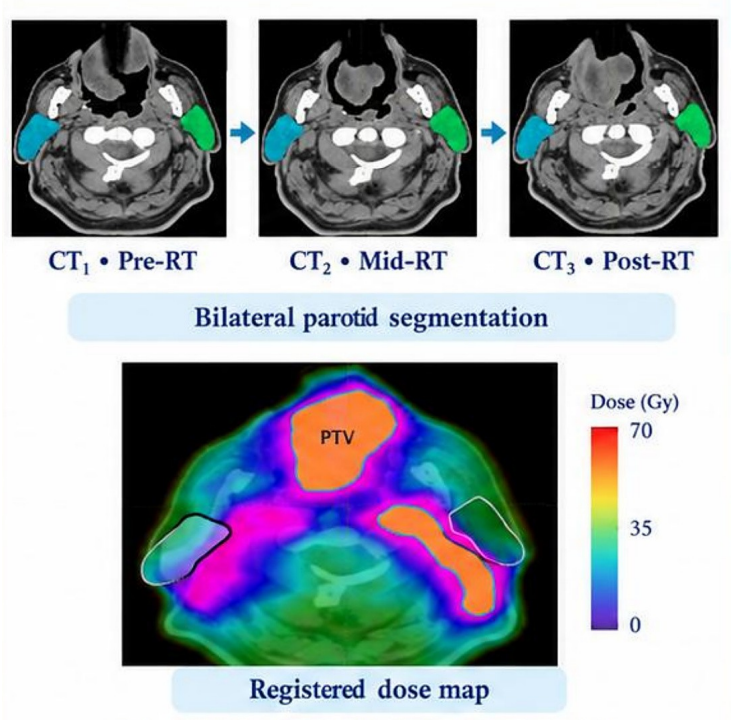
Key points

- Classify lesion type from CT
- Five tumour types vs normal
- Support for, not a replacement of, biopsy

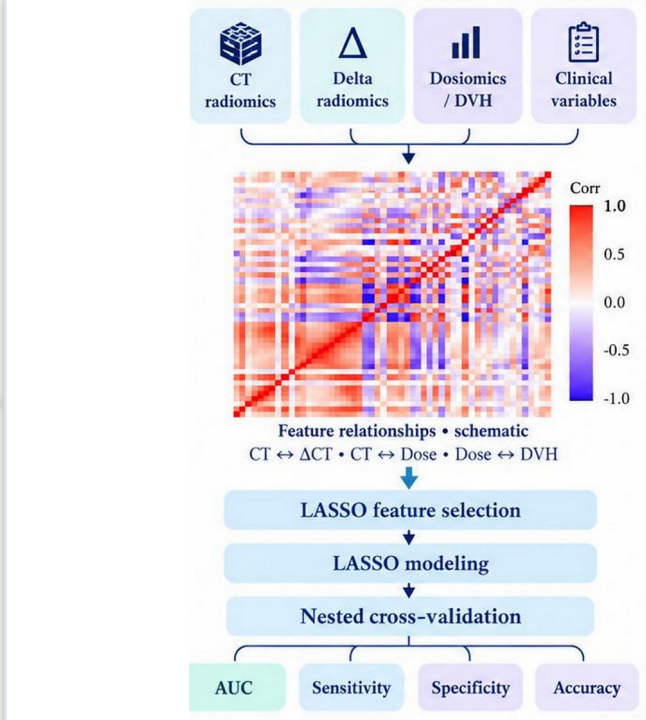
A virtual biopsy alongside tissue diagnosis

RADIOMICS · TOXICITY PREDICTION

Delta radiomics and dose for late xerostomia



Parotid CT before, during, after RT



CT, delta, dose and clinical features

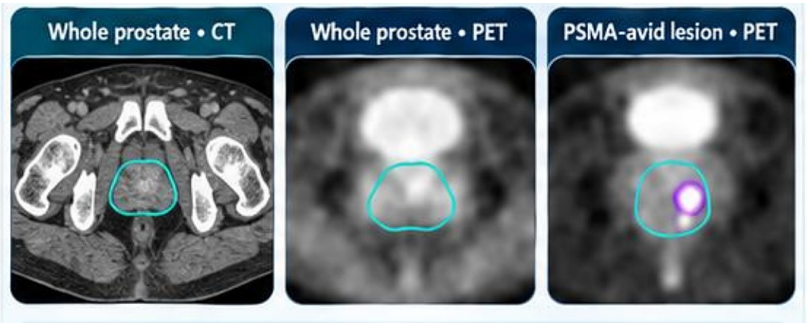
Key points

- Parotid changes during RT matter
- Delta radiomics + dose predict best
- Supports toxicity-aware planning

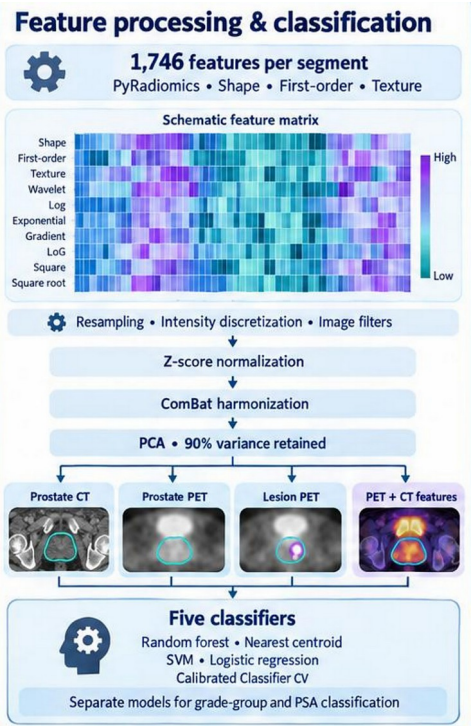
Longitudinal imaging and dose predict dry mouth

RADIOMICS · PSMA PET/CT

Multicentre PSMA PET/CT radiomics for risk



PSMA PET/CT, 243 patients, 4 centres



Harmonized features, five classifiers

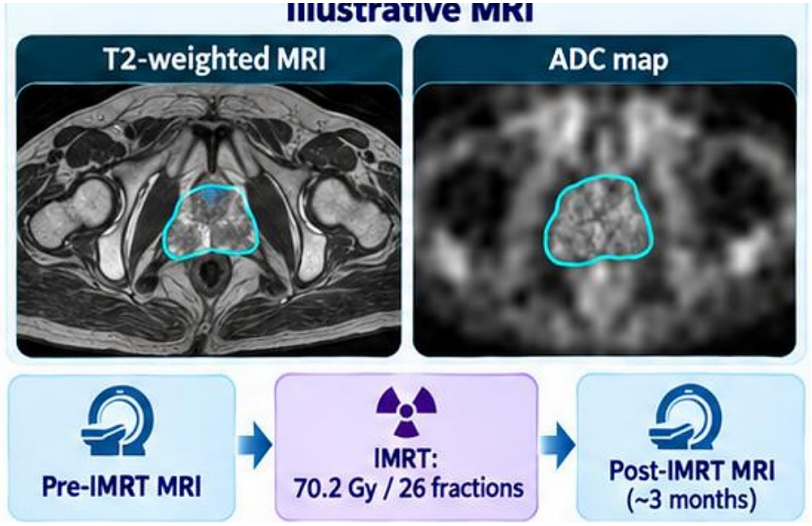
Key points

- Predicts Gleason group and PSA level
- Tested on external centres
- Lesion PET and PET/CT fusion best

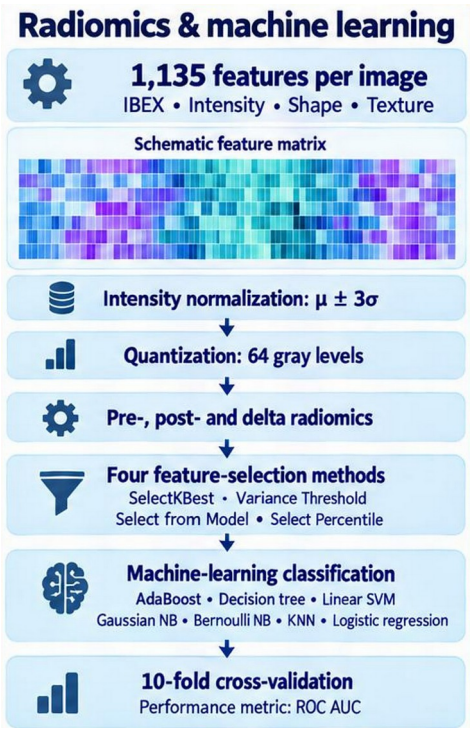
Non-invasive risk stratification from PSMA PET/CT

RADIOMICS · TREATMENT RESPONSE

MRI radiomics for IMRT response in prostate



T2 MRI and ADC before and after IMRT



1,135 features, seven classifiers

Key points

- Pre-treatment MRI predicts response
- Also classifies Gleason and stage
- Exploratory cohort (33 patients)

MRI radiomics for response and tumour characterization

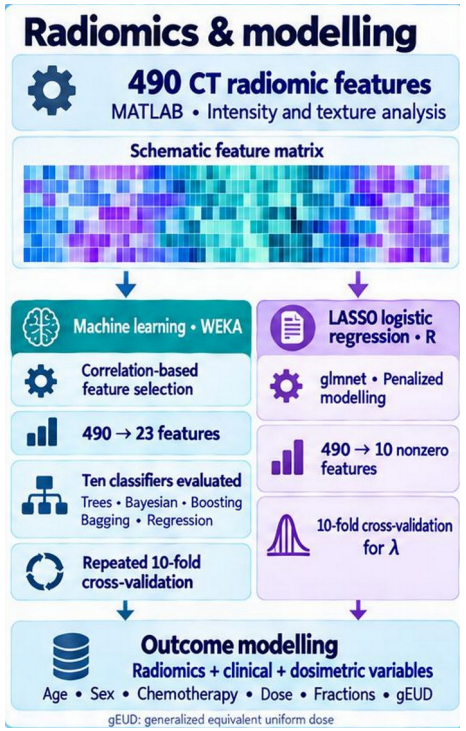
Abdollahi et al., La Radiologia Medica 2019

COCHLEAR CT RADIOMICS · TOXICITY

Predicting radiation-induced hearing loss



Cochlear CT, 47 patients



490 features, ML and LASSO

Key points

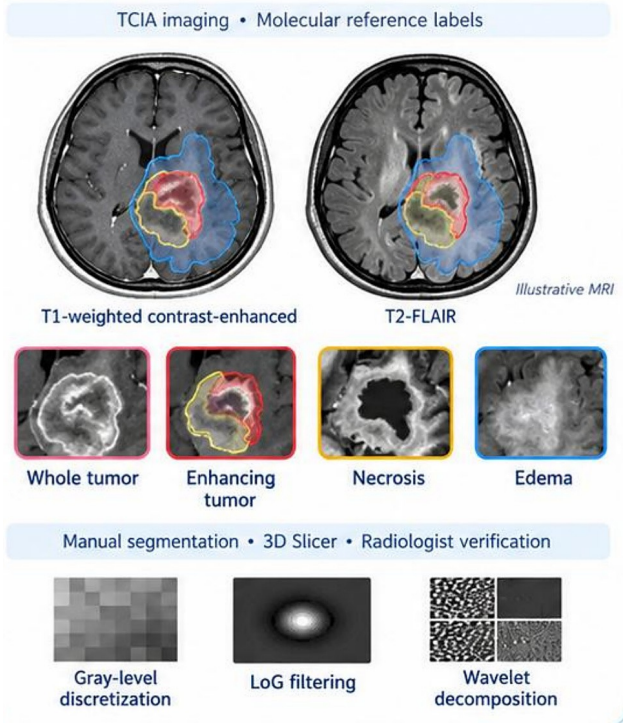
- About 40% of ears lost hearing
- Radiomics + dose (gEUD) predict risk
- Small cohort; needs validation

Imaging the cochlea to predict hearing loss

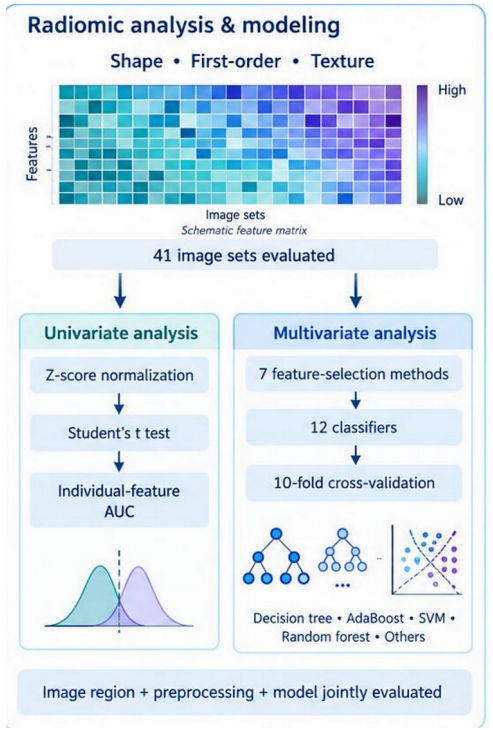
Abdollahi et al., Physica Medica · doi:10.1016/j.ejmp.2017.10.008

RADIOGENOMICS · GLIOBLASTOMA

MRI radiomics for MGMT promoter methylation



MRI tumour sub-regions, 82 patients



41 image sets, 12 classifiers

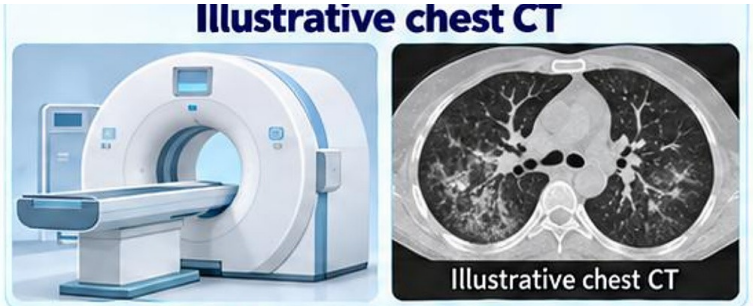
Key points

- Imaging signature of a gene marker
- Edema region most informative
- Internal validation only

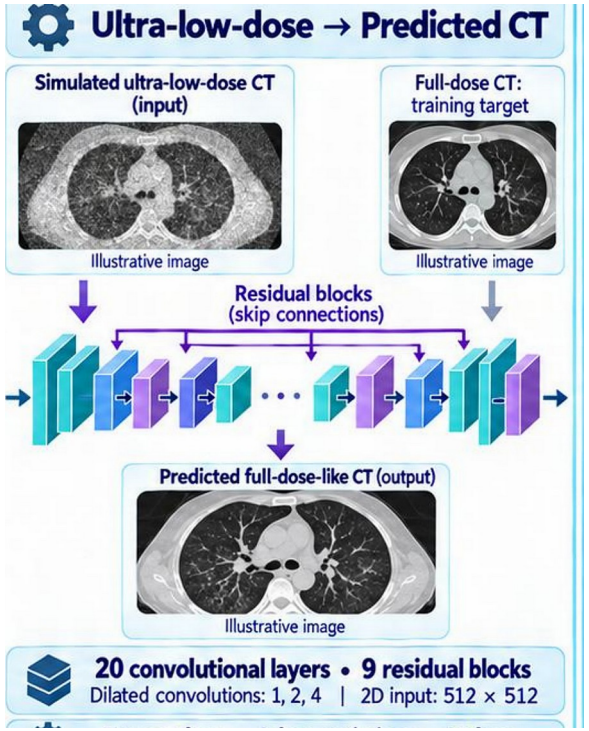
Non-invasive MGMT status from MRI

DEEP LEARNING · IMAGE QUALITY

Deep learning for ultra-low-dose chest CT



1,141 chest CTs, 9 centres



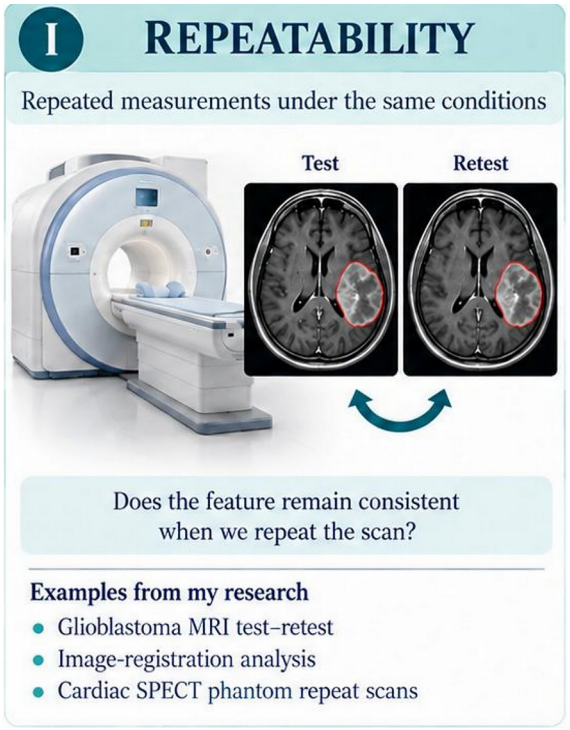
Residual network: low → full dose

Key points

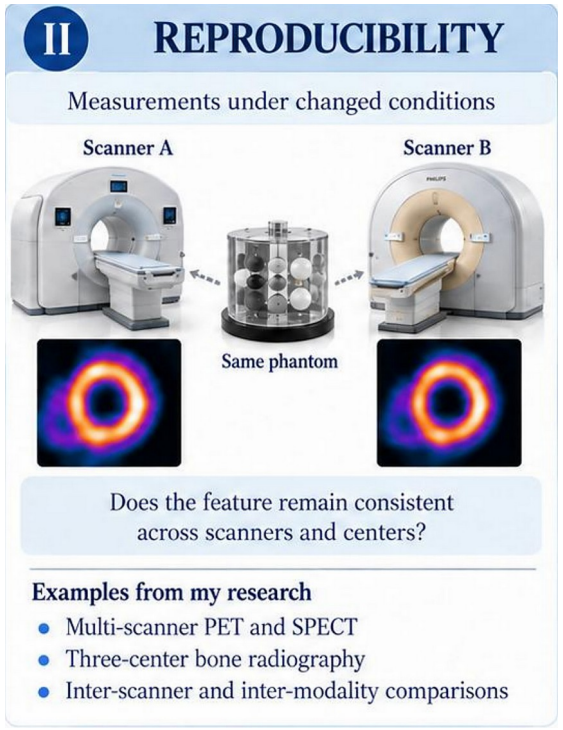
- Up to 89% less radiation
- Image quality near full dose
- Lesion fidelity must be checked

Lower dose with preserved image quality

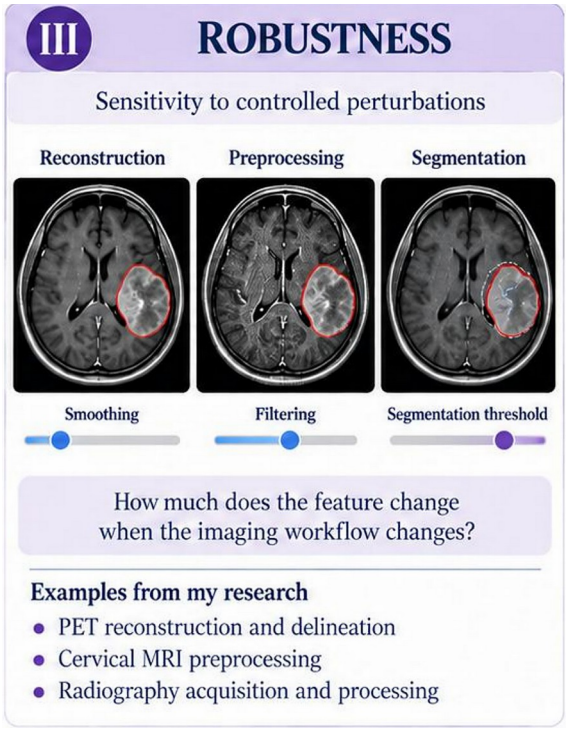
Repeatability, reproducibility and robustness



Repeatability: same conditions



Reproducibility: across scanners

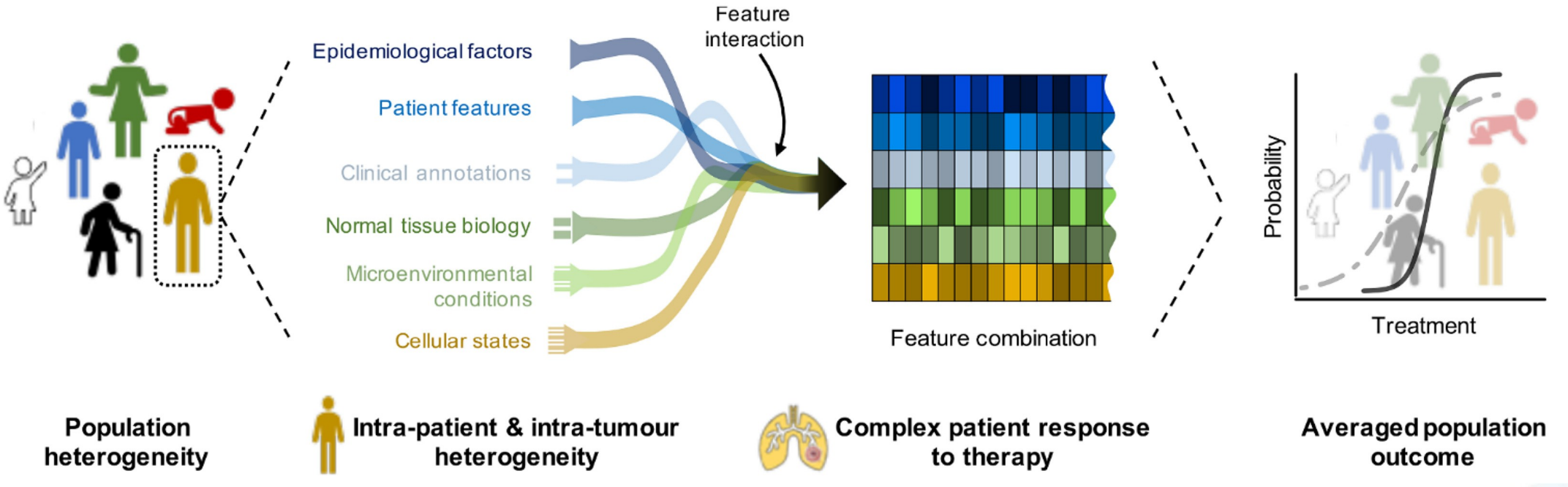


Robustness: workflow changes

Goal: find stable features and the sources of variability

THE CHALLENGE

Treatment is designed for the average patient



Averaging hides the individual patient

THE SOLUTION · DIGITAL TWINNING

Trials for patients who are not average



iScience

CellPress
OPEN ACCESS

Perspective

Designing clinical trials for patients who are not average

Thomas E. Yankeelov,[1,2,3,4,5,9,*] David A. Hormuth II,[4,5] Ernesto A.B.F. Lima,[4,6] Guillermo Lorenzo,[4,10] Chengyue Wu,[4] Lois C. Okereke,[4] Gaiane M. Rauch,[7] Aradhana M. Venkatesan,[7] and Caroline Chung[8]

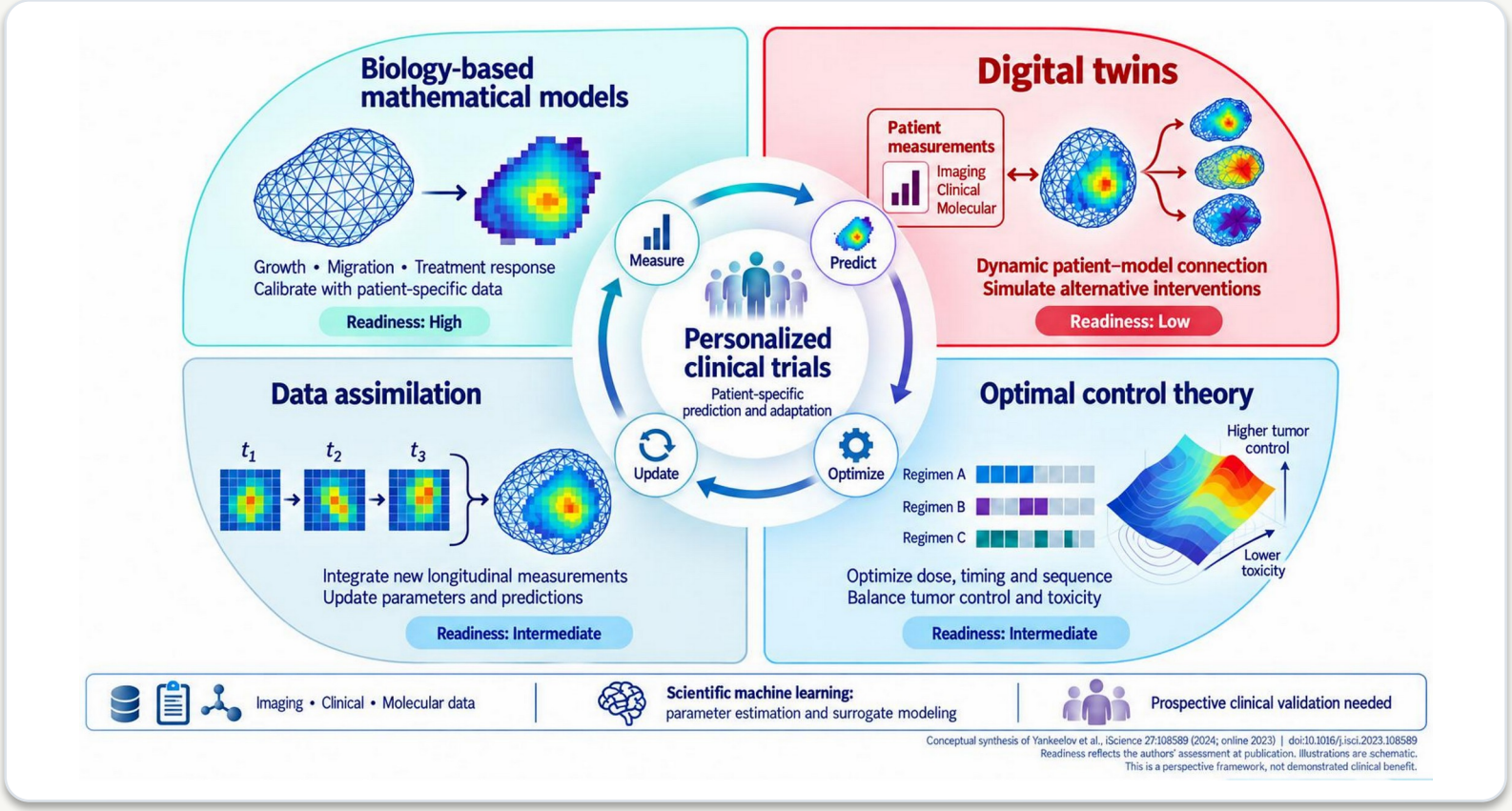
The idea

- Model each patient, not the average
- Calibrate with their own data
- Update as treatment proceeds

From averaged evidence to personalized, adaptive decisions

THE SOLUTION · DIGITAL TWINNING

Four building blocks for personalized trials



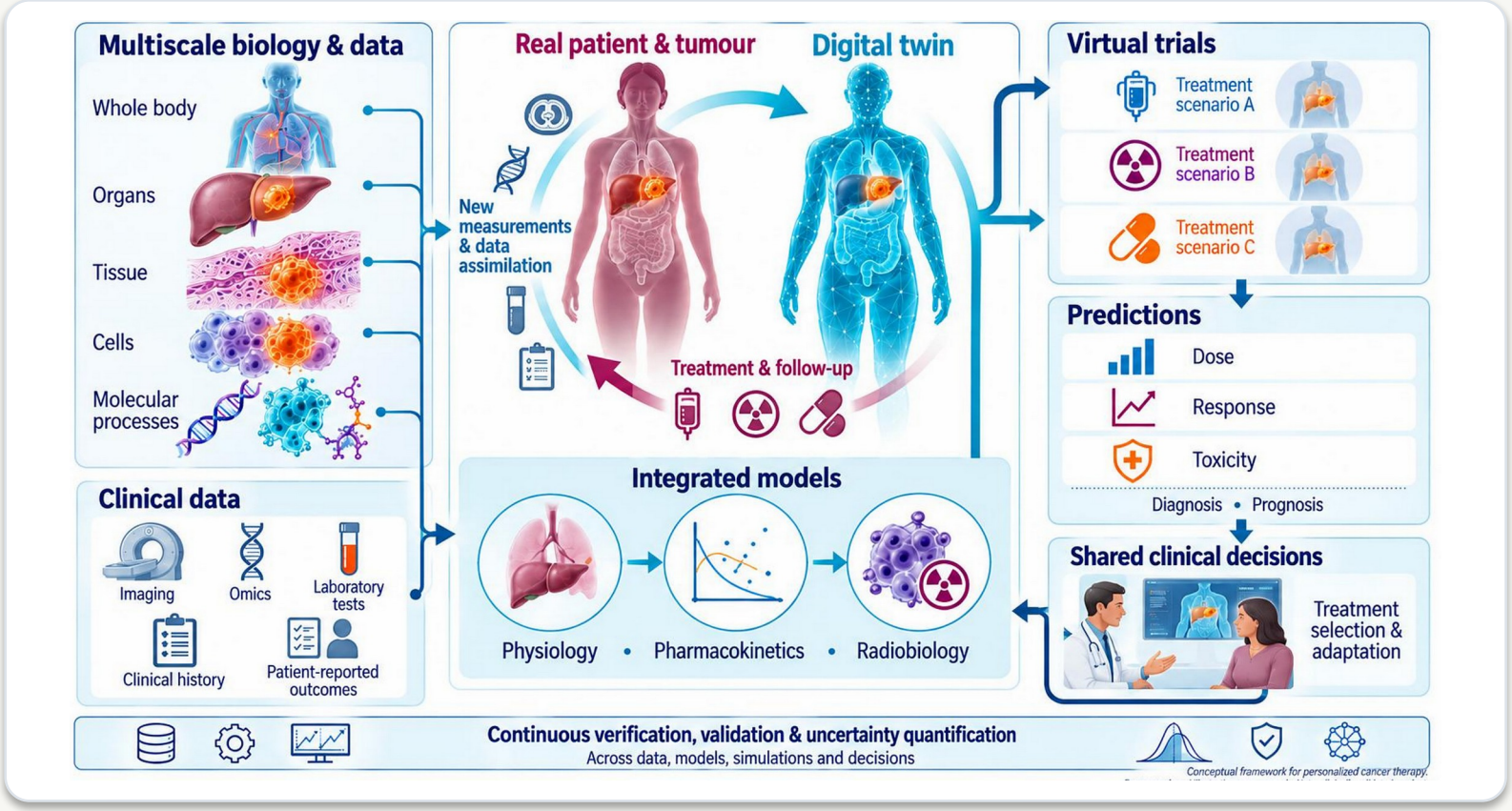
Building blocks

- Biology-based models
- Data assimilation
- Optimal control
- Digital twins

Measure → predict → optimize → update

THE SOLUTION · DIGITAL TWINNING

What is a digital twin?



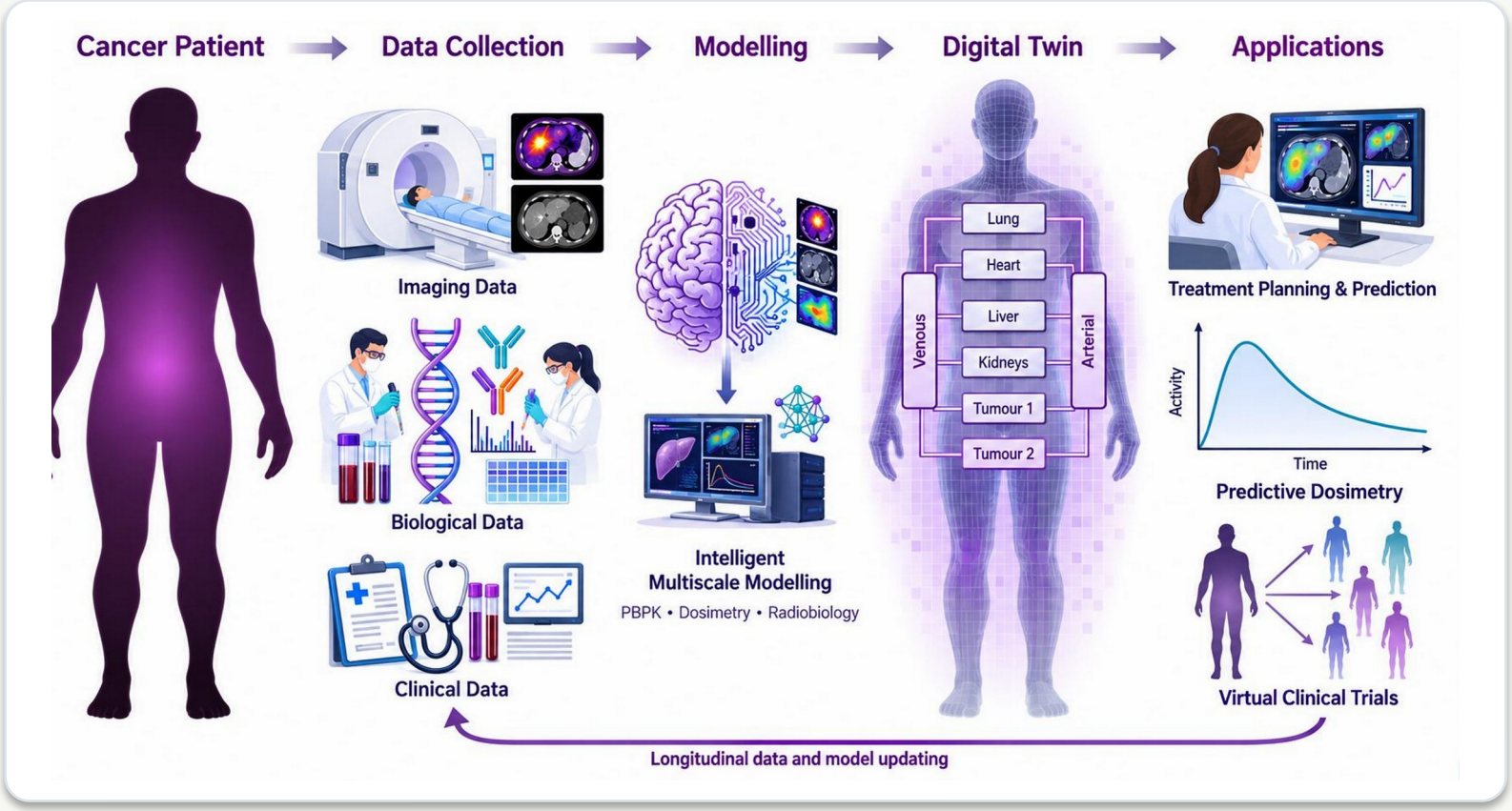
Definition

- A virtual copy of one patient
- Updated with that patient's data
- Predicts and guides decisions

A two-way link between patient and model

THERANOSTIC DIGITAL TWINNING

From the patient to a theranostic digital twin



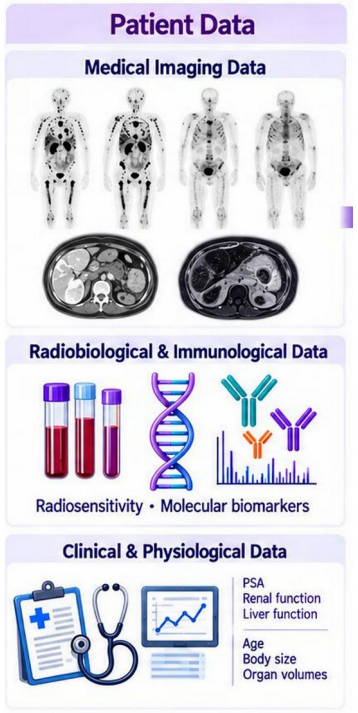
Steps

- Imaging, biological, clinical data
- PBPK, dosimetry, radiobiology
- Planning and virtual trials

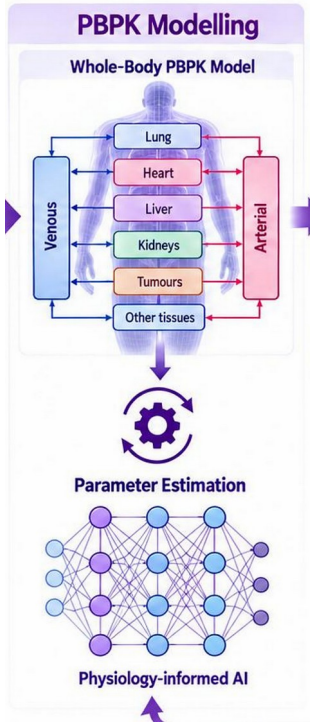
Longitudinal data keep updating the twin

THERANOSTIC DIGITAL TWINNING

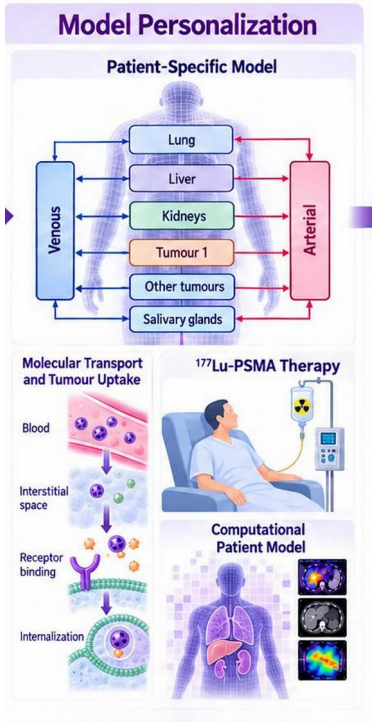
Building the twin: PBPK model for ^{177}Lu-PSMA



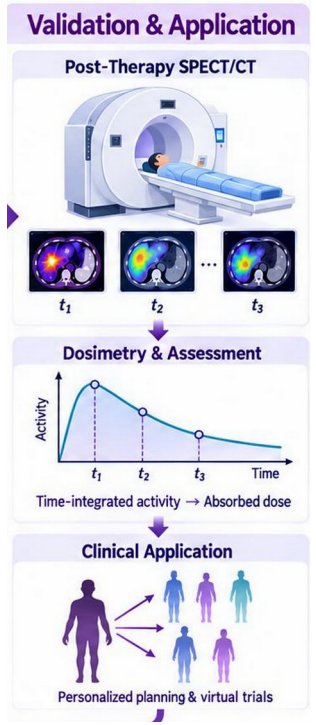
Patient data



PBPK model



Personalization

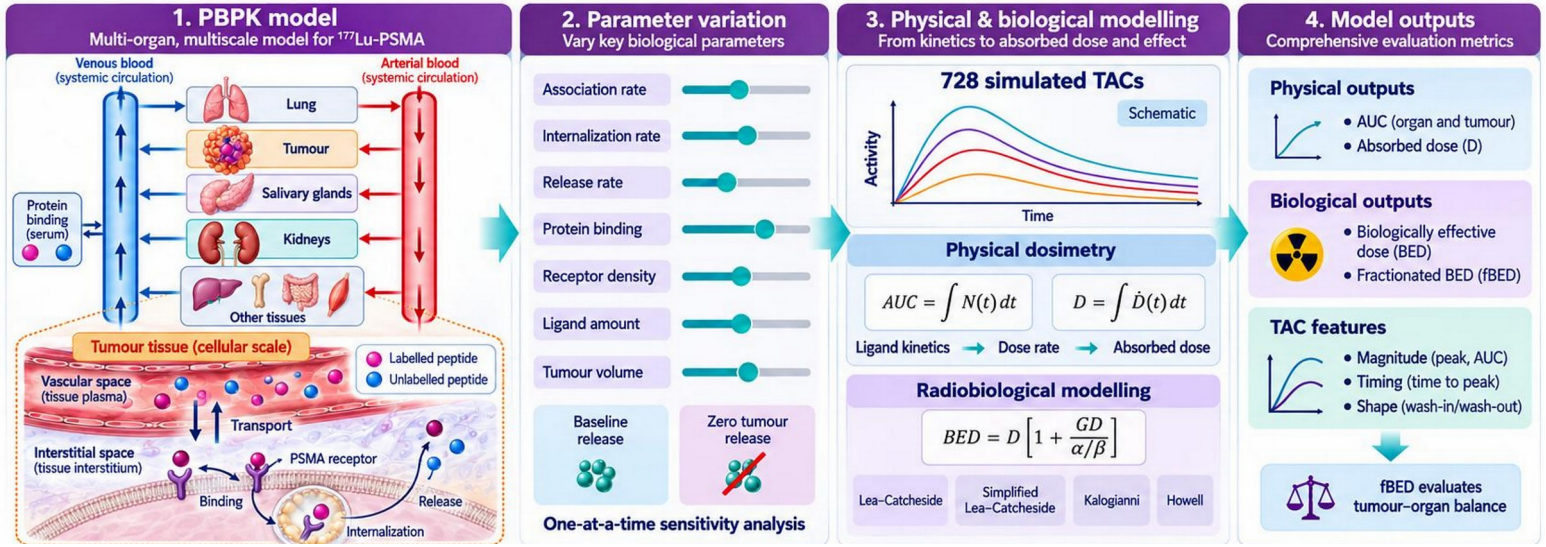


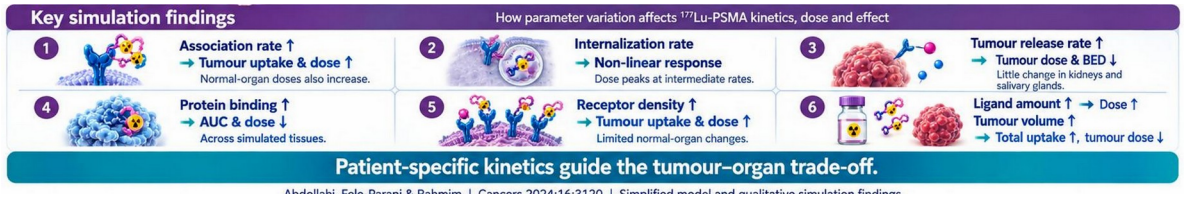
Validation & use

Model refinement with longitudinal data closes the loop

THERANOSTIC DIGITAL TWINNING

Model-informed ^{177}Lu-PSMA optimization

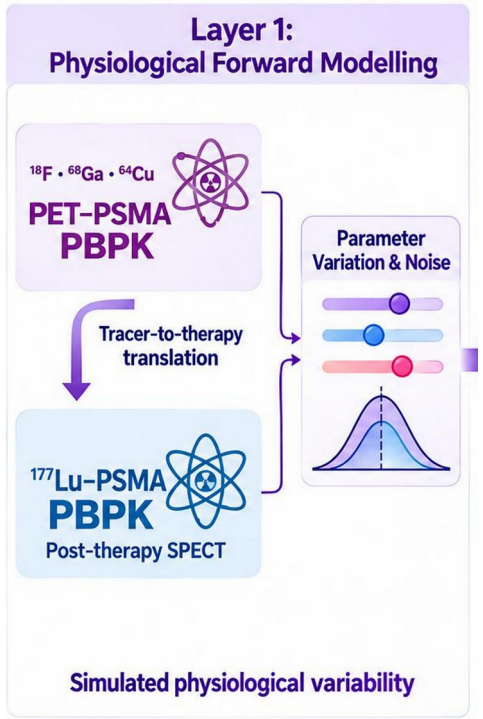




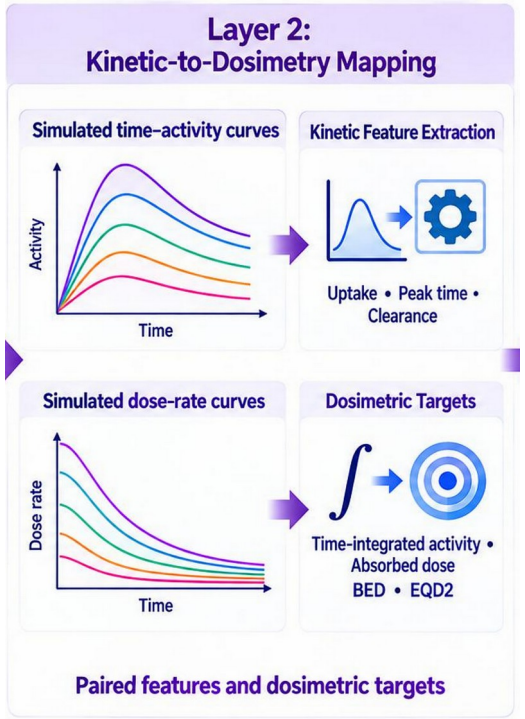
Patient-specific kinetics guide the tumour–organ trade-off

APPLICATION · PREDICTIVE DOSIMETRY

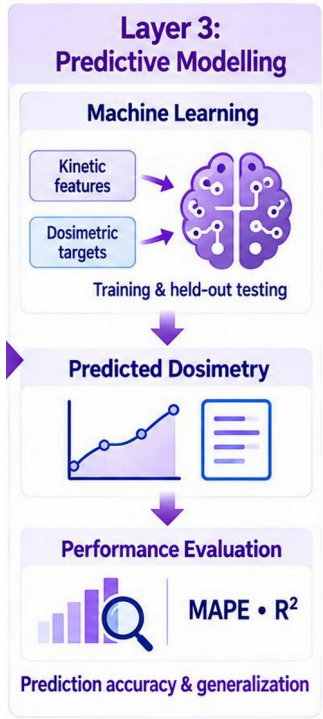
Predicting ^{177}Lu dosimetry from PET kinetics



Layer 1: physiological modelling



Layer 2: kinetics → dosimetry



Layer 3: machine learning

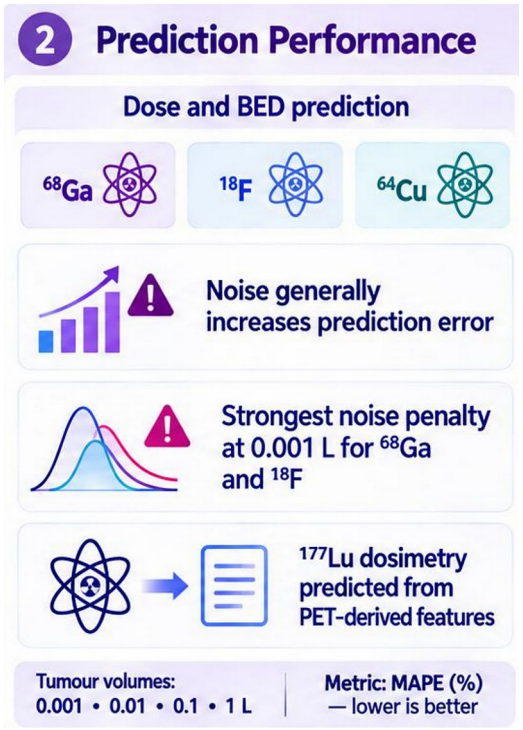
Pre-therapy PET kinetics → predicted therapy dose

APPLICATION · PREDICTIVE DOSIMETRY

Predictive dosimetry: key findings



Peak uptake (AMax) leads



Noise matters most in small tumours



Model interpretation (SHAP)

Uptake magnitude and kinetics support dosimetry prediction